

Project: Food Remedy API

Ticket: RE-024 – Mobile Application Data Validation and Integrity Best Practices

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Table of Contents

1. Introduction
2. Food and Product Data Validation
3. Handling Common Data Quality Problems
4. Validating External API Data
5. Client-Side, Backend and Database Validation
6. Maintaining Data Integrity When Information Changes
7. Recommendations for Food Remedy
8. Conclusion
9. References

1. Introduction

To give users information about ingredients, nutrition, and allergies, Food Remedy uses food and product data. It is crucial to examine the data for missing values, wrong formats, duplication, and inconsistencies before it is displayed or utilized by application features.

This is especially crucial when data originates from outside sources like Open Food Facts, where the accuracy and completeness of product records can differ. Errors and warnings are examples of data-quality indicators found in Open Food Facts' product data (Open Food Facts, 2026).

In order to increase the accuracy of food information prior to public distribution, this study explores useful validation methods that Food Remedy may employ. It focuses on handling partial or ambiguous information, external API data, food-specific validation, validation at several application layers, and preserving consistency when data is modified.

2. Food and Product Data Validation

Validation ought to verify more than just a field's existence. Additionally, data must be of the appropriate kind, format, unit, and reasonable value. Both syntactic validation—which determines whether the data has the anticipated structure—and semantic validation—which determines whether the value makes sense in its context—are advised by OWASP (OWASP Foundation, 2026).

For instance, "protein": -15 should still fail validation since a negative protein quantity is meaningless, even though "protein": 15 may have the correct numeric format.

Table 1. Recommended Food Data Validation

Problem	Example	Recommended Handling
Missing data	Ingredients unavailable	Mark information as unavailable
Empty data	ingredients = ""	Treat as missing
Incorrect format	Protein value contains text	Reject affected value
Invalid value	Negative nutrition quantity	Flag as invalid
Duplicate data	Same barcode appears twice	Reconcile using identifier/version
Unknown unit	Energy value has no unit	Do not perform calculation
Conflicting data	Ingredients indicate milk but allergen field does not	Mark as uncertain
Outdated data	Newer product information exists	Revalidate and replace/update
Missing allergen field	No allergen information available	Treat as unknown, not allergen-free

Nutrition values are broken down by unit and whether they apply per serving or per 100 g in Open Food Facts. Instead of treating all nutrition values equally, Food Remedy should maintain these distinctions (Open Food Facts, 2026).

Additionally, missing nutrition data shouldn't be set to zero. For instance, 0 g sugar indicates something different than the lack of sugar information.

3. Handling Common Data Quality Problems

Food Remedy may come into information that is inconsistent, redundant, inaccurately presented, or incomplete. Not every one of these circumstances should be addressed in the same manner.

Table 2. Common Data Problems and Suitable Handling

Problem	Example	Recommended Handling
Missing data	Ingredients unavailable	Mark information as unavailable
Empty data	ingredients = ""	Treat as missing
Incorrect format	Protein value contains text	Reject affected value
Invalid value	Negative nutrition quantity	Flag as invalid
Duplicate data	Same barcode appears twice	Reconcile using identifier/version
Unknown unit	Energy value has no unit	Do not perform calculation
Conflicting data	Ingredients indicate milk but allergen field does not	Mark as uncertain
Outdated data	Newer product information exists	Revalidate and replace/update
Missing allergen field	No allergen information available	Treat as unknown, not allergen-free

Food Remedy may come into information that is inconsistent, redundant, inaccurately presented, or incomplete. Not every one of these circumstances should be addressed in the same manner.

Data classification into four states would be a helpful strategy for Food Remedy:

Valid: all necessary checks have been completed.

Incomplete: While some information is absent, it is still useful.

Uncertain: Conflicting or unclear information makes it difficult to interpret with confidence.

Information that doesn't meet crucial validation criteria is considered invalid.

This is especially crucial for information on allergens. For packaged food notifications in Australia and New Zealand, FSANZ mandates specific allergy wording (FSANZ, 2026). Food Remedy should keep the original source information for traceability, but it can normalize accepted terms for matching purposes.

Food Remedy should refrain from drawing a firm safety determination if allergy information is incomplete or contradictory.

For instance:

Preferred: "Allergen information could not be verified."

Avoid: “No allergens detected.”

when the source data is not complete.

4. Validating External API Data

Before become reliable Food Remedy data, information obtained from an API or external database needs to be verified.

A successful API request does not verify that every product field is complete or appropriate for Food Remedy; it merely verifies that data was returned.

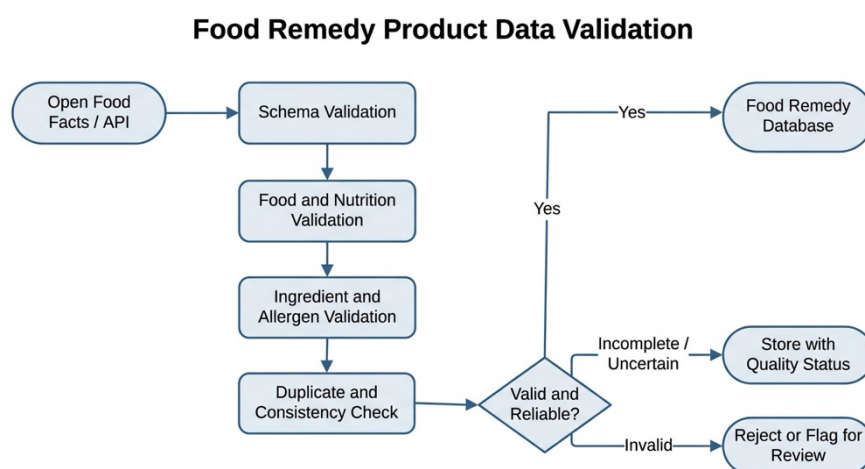
An actual validation procedure would be:

External API → Schema Check → Field Validation → Normalisation → Consistency Check → Approved Data

The application should verify:

- if parsing the API response is possible;
- the presence of anticipated fields and data types;
- the validity of product identifiers;
- the usability of nutrition values and units;
- The interpretability of information on ingredients and allergens;
- if there are already duplicate records; • whether significant fields clash; and
- If the data is up to date enough.

The product schema of Open Food Facts is likewise subject to change over time and may include new fields. Therefore, while identifying missing data that the application truly relies on, Food Remedy should safely disregard unsupported extra fields (Open Food Facts, 2026).



5. Client-Side, Backend and Database Validation

There should be several layers of validation. Although client-side validation is helpful for quick feedback, client-side checks can be circumvented, thus it shouldn't be the main safeguard. Authoritative input validation should be carried out on the server side, according to OWASP (OWASP Foundation, 2026).

Table 3. Validation Approach Comparison

Area	Client-Side	Backend/API	Database
Immediate user feedback	Strong	Moderate	Weak
Basic format checking	Strong	Strong	Moderate
External API validation	Limited	Strong	Weak
Nutrition validation	Basic	Strong	Basic
Allergen logic	Display checks	Strong	Store validated result
Business rules	Limited	Strong	Limited
Duplicate prevention	Preliminary	Strong	Strong
Structural integrity	Limited	Strong	Strong
Can be bypassed by client	Yes	Much harder	No for enforced constraints
Recommended role	Early checks	Main validation layer	Final integrity protection

The primary food data validation for Food Remedy should be done by the Backend/API. While the database should safeguard structural integrity and stop invalid or unauthorised writes, the mobile client should avoid blatantly incorrect input and suitably present incomplete information.

Firestore Security Rules offer an extra database-level control by verifying suggested document modifications and limiting writes according to predetermined criteria (Google, 2026).

6. Maintaining Data Integrity When Information Changes

Validation shouldn't only take place when a product is first introduced because food information can change over time.

When a product is changed, the new data should go through the same validation procedure once more.

Table 4. Data Integrity Practices

Practice	Food Remedy Application
Stable product identifier	Use barcode to identify existing products

Update timestamp	Record when product information changed
Source information	Record where data originated
Dataset/version information	Identify which dataset contains the product
Duplicate checking	Prevent multiple unintended records for one barcode
Revalidation	Validate changed products again
Conflict handling	Prefer approved authoritative information
Validation status	Store valid/incomplete/uncertain/invalid state
Logging	Record important validation failures
Quarantine	Prevent critically invalid records reaching users

These controls would also make it simpler to look into the reasons for a specific product's acceptance, rejection, or limited display.

7. Recommendations for Food Remedy

Instead of depending solely on checks within the mobile interface, Food Remedy should incorporate validation as part of its product-data flow, according to the report.

Table 5. Recommended Actions

Priority	Recommendation	Reason
High	Validate external product data before it is used	External data can be incomplete or inconsistent
High	Make Backend/API validation authoritative	Client validation alone can be bypassed
High	Use valid, incomplete, uncertain and invalid states	Prevents unreliable information appearing definite
High	Never convert missing nutrition values to zero	Missing and zero have different meanings
High	Treat missing allergen information as unknown	Prevents unsupported allergen-safety conclusions
High	Validate nutrition units and measurement basis	Prevents incorrect calculations
High	Revalidate products when information changes	Previously valid data can become outdated
Medium	Normalise allergen terminology while retaining original text	Improves matching while preserving traceability
Medium	Detect duplicate products using stable identifiers	Improves catalogue consistency
Medium	Store source and update information	Supports freshness and troubleshooting

Medium	Use database-level integrity/security controls	Adds another validation layer
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Thus, the following would be an appropriate Food Remedy validation flow:

External data → Backend validation → Quality status → Database → Mobile application

For features that need accurate product information, only data that has passed the necessary tests should be used. Information that is unclear or incomplete can still be helpful, but the application should properly convey its limits.

8. Conclusion

Before going public, Food Remedy needs accurate food, nutrition, and allergen information. The mere fact that data was successfully acquired from other sources does not inevitably make it legitimate. Product identifiers, necessary fields, nutrition values and units, ingredients, allergies, duplication, and consistency between related fields should all be checked during validation. It's important to identify verified information from missing, partial, and ambiguous information. The best method is multi-layer validation, in which the database offers additional structural and integrity controls, the backend/API handles the primary food-data validation, and the mobile application handles basic checks and safe presentation. In particular, Food Remedy should refrain from considering missing values to be confirmed. Missing allergen information should not be construed as allergen-free, nor should missing nutrition data become zero. To reduce the chances of wrong information of the product shown to the user or used by any feature of Food Remedy API, these practices should be implemented which will improve reliability of application.

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